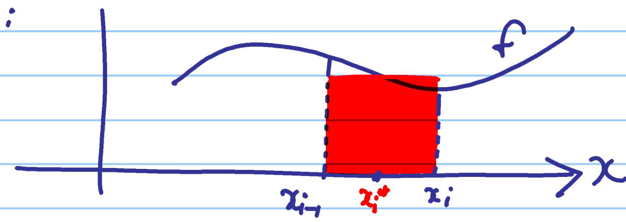
Note Title 5/11/2018

Unit 5.3 The definite integral. (p.378)

We start with the general Riemann sum (see p.379) for an area:



Method:

- We divide $[a,b]$ into n sub-intervals.
- The width of each sub-interval is $\Delta x = \frac{b-a}{n}$.
- Consider $[x_{i-1}, x_i]$
- Pick a random point: $x_i^* \in (x_{i-1}, x_i)$
- Riemann sum: $\sum_{i=1}^{n} f(x_i^*)\Delta x$

And the definite integral is:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

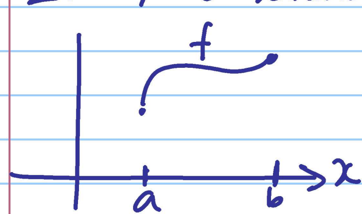
The definite integral: $\int_a^b f(x)\,dx$

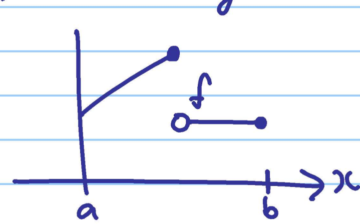
a, b: limits of integration

$f(x)$ → integrand

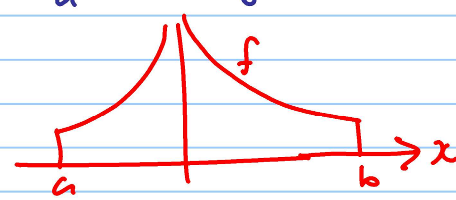
Theorem: If f is continuous on $[a,b]$ or if f has a finite number of discontinuities, then f is integrable.

Examples of functions that are integrable:





The following function is not integrable on $[a,b]$:



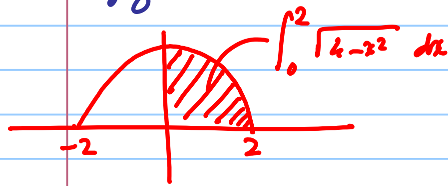
Properties of the definite integral: NB

See p. 385, 387.

You <u>must</u> know and be able to apply <u>all</u> these properties. (8 properties!)

Examples: Calculate without using anti-derivatives.

1) $\int_0^2 \sqrt{4-x^2}\, dx = \frac{1}{4}(4\pi) = \pi.$



Area of circle with radius 2: $\pi r^2 = 4\pi$.

2) $\int_0^2 -\sqrt{4-x^2}\, dx = -\pi$

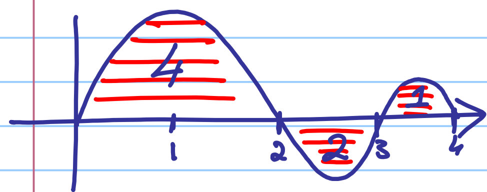
Area is positive, but integral is negative.

3) $\int_a^b (-3)\, dx =$

4) $\int_{-1}^2 2x\, dx =$

Please see next page for more examples:
It is important to know the difference between "calculating an area" and "find the integral".

5) ## Area and integrals:



$$\int_0^3 f(x)\,dx = 4-2 = 2$$

$$\int_2^4 f(x)\,dx = -2+1 = -1$$

$$\int_0^4 f(x)\,dx = 4-2+1 = 3$$

What is the total area enclosed by the graph and the x-axis between $x=0$ and $x=4$?

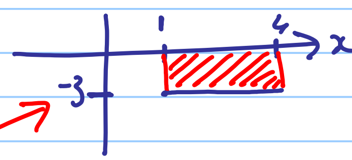
$$4+2+1 = 7$$

The Fundamental Theorem of Calculus: (P.396)

$$\int_a^b f(x)\,dx = F(b)-F(a), \text{ where } F'(x) = f(x).$$

Examples:

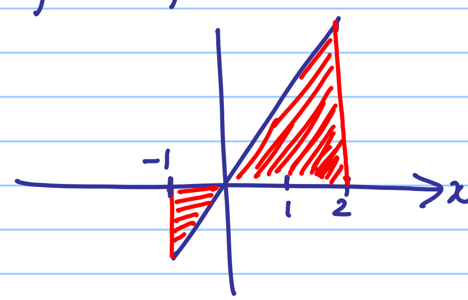
1) $\int_0^4 (x^2+4)\,dx$
2) $\int_{-1}^2 e^{-x}\,dx$
3) $\int_{\pi/2}^{2\pi} \cos x\,dx$



Solutions:

3) $\int_1^4 (-3)\,dx = (-3)(4-1) = -9$

4) $\int_{-1}^{2} 2x\,dx$



$$= \int_{-1}^{0} 2x\,dx + \int_0^2 2x\,dx$$

$$= x^2\Big|_{-1}^{0} + x^2\Big|_0^2$$

$$= (0-1)+(4-0)$$

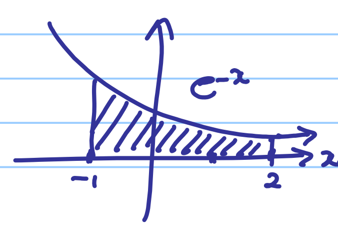
$$= -1+4 = \underline{3}$$

More examples:

① $\int_0^4 (x^2+4)\,dx = \left[\frac{1}{3}x^3 + 4x\right]_0^4 = \frac{64}{3} + 16$

② $\int_{-1}^{2} e^{-x}\,dx = -e^{-x}\Big|_{-1}^{2}$

$$= -e^{-2} + e^{1} = e - \frac{1}{e^2}$$



(8) $\int_{\pi/2}^{2\pi} \cos x \, dx = \sin x \Big|_{\pi/2}^{2\pi} = \sin 2\pi - \sin \frac{\pi}{2}$

$= 0 - 1$

$= -1.$